SURAJ SURESH

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SUMMARY

M.S. Data Science student (4.0 GPA, expected Spring 2028) seeking Summer 2027 Machine Learning/AI Engineering, Data Engineering, or Infrastructure internships. A year of ML work in fintech: fine-tuned a transformer (PyTorch, Hugging Face) that shipped to production users, built the Python/SQL pipelines behind it, and owned its GCP-to-AWS migration. Since then, built three AI systems end to end: a benchmarked RAG system, a vision-language search system, and a deployed multimodal assistant, each with CI and documented design decisions.

EXPERIENCE

Machine Learning Intern **June 2023 – May 2024**
FitBUX (financial technology) Austin, TX (Remote)

- Fine-tuned BERT (PyTorch, Hugging Face Transformers) on a financial-text corpus and shipped its embeddings as the semantic-matching layer over a 1,000-pair advisor-validated Q&A bank, powering a conversational guidance feature for production users.
- Built the Python and SQL data pipelines that assembled the 1,000+ advisor-validated records behind the model.
- Owned production deployment end to end: migrated the service from GCP Vertex AI to AWS EC2 and S3 and stood up CloudWatch monitoring for the live service.

PROJECTS

Agentic RAG System over LLM Inference-Optimization Literature — LangGraph, Qdrant, RAGAS, FastAPI, Docker, CI **Jun–Jul 2026**

- Evaluated dense, hybrid (BM25 + embedding), and cross-encoder-reranked retrieval on a 300-question factoid benchmark (rag-mini-wikipedia); hybrid lifted MRR from 0.594 to 0.639 (Hit@5 0.697 to 0.717) and shipped as the default.
- Traced reranking's MRR drop (0.639 to 0.520) to the answer-containment metric rewarding literal answer strings over relevance; disabled it by default and recorded the trade-off and follow-up plan as an architecture decision record.
- Rebuilt the agent on LangGraph after a control test traced failed tool calls to the prior framework's middleware, not the 14B model.
- Measured generation quality with a NaN-tolerant RAGAS harness (local judge by default, hosted-judge fallback), tracked in W&B.

Explainable Multimodal Retrieval (Person of Interest) — SigLIP-2, FAISS, Qwen2.5-VL, SLURM **Apr–May 2026**

- Architected a describe-then-retrieve search system (SigLIP-2 shared text-image embeddings, FAISS IVF-Flat cosine retrieval, Qwen2.5-VL explaining each match in query context) and validated it end to end on a synthetic CelebA-format corpus.
- Designed a pre-registered encoder ablation (CLIP-ViT-L/14 vs. SigLIP-2 base vs. large) on CelebA with hypothesis and metrics fixed before the run, packaged as SLURM jobs for a shared GPU cluster; real-model run pending.

LifeLens — Multimodal Assistant (deployed) — React 18, FastAPI, multimodal LLM APIs, pytest, CI, Vercel **Jun–Aug 2026**

- Designed and deployed a cross-platform PWA that reads photos of documents, appliances, and labels and answers open questions with cited web sources; every model response is contracted to a strict JSON schema and Pydantic-validated server-side.
- Hardened the API: a FastAPI proxy keeps the model key off the client and enforces input validation, payload caps, and rate limits.
- Fixed a silent failure where full-size phone photos exceeded the vision API's payload limit by downscaling images client-side.

TECHNICAL SKILLS

Languages: Python, SQL, R, C++, JavaScript
Machine Learning: scikit-learn, PyTorch, Hugging Face Transformers, transformer fine-tuning, text and multimodal embeddings
Evaluation & Experiments: retrieval benchmarking, controlled ablations, retrieval metrics (MRR, recall@k), RAGAS, Weights & Biases
LLM & Agent Frameworks: LangChain, LangGraph, RAG pipeline design, agents, tool calling, MCP, local inference (Ollama)
Search & Vector Databases: Qdrant, FAISS, dense and hybrid (BM25 + embedding) retrieval, cross-encoder reranking
Engineering & Infrastructure: FastAPI, Pydantic, React 18, Docker, CI/CD, AWS (EC2, S3, CloudWatch), GCP (Vertex AI, GCS), SLURM
Relevant Coursework: Machine Learning Economics, Structured Query Language (SQL), Python Programming, Data Structures & Algorithms, Dynamic Programming & Greedy Algorithms, Applied Discrete Math, Computing Fundamentals 1 & 2

EDUCATION

M.S. in Data Science, University of Colorado, Boulder, GPA: 4.0/4.0 Expected Spring 2028
B.A. in Business Management Economics, University of California, Santa Cruz June 2025

TRAINING

- **Become an AI Engineer, ByteByteGo (Jun–Aug 2026):** six projects incl. an evaluated RAG chatbot and an MCP tool-calling agent
- **Large Language Models Bootcamp, SupportVectors (Jan–Mar 2026):** 24 labs and a team capstone across RAG, fine-tuning, agents, and security.